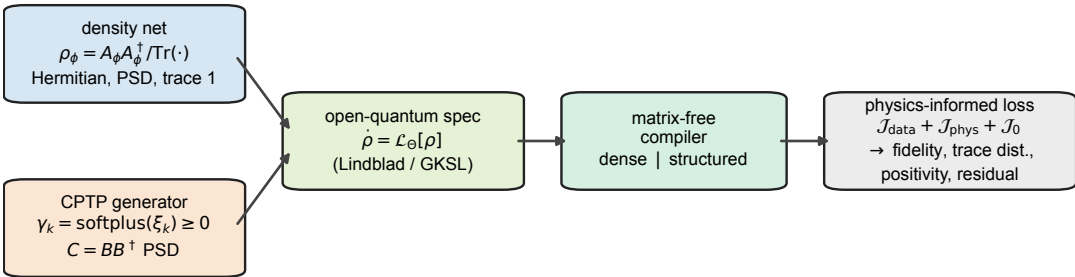


structure-preserving residual PINN for open quantum dynamics



dense ↔ structured residual agree to floating-point precision • physicality guaranteed at every step (Theorem 1)